

Interview Questions for the Blockchain-Based

Design Pattern Decision Model

Part I. Demographic information

1. Do you usually focus only on academic research, or also deeply involved in some industrial projects in practice?
2. When did you start to study software architecture and design patterns? And when did you get involved in the blockchain field?
3. What are your views on blockchain design patterns and decision models?

You can consider the current state of research, contributions to the industry, future development, etc.

Part II. Completeness Evaluation

This part of the question focuses on researchers working on blockchain research within the academic community.

4. Is the content of the decision model sufficient? Are there elements that need to be added, such as design patterns, quality attributes, etc.?

Examples of aspects that can be considered:

- Is the design pattern content comprehensive enough?
- Are the quality attributes involved comprehensively?
- Are there additional elements that need to be added to the model?

5. What do you think of the structure of our decision model? What is its level of coverage for blockchain system architecture?

Examples of aspects that can be considered:

- Is the consideration of the blockchain hierarchy (e.g., six-layer structure) comprehensive?
- Is the identification of the content of each concern aspect complete?
- Are there any layers that need to be extended?

6. What do you think about the uneven distribution of design patterns across layers (e.g. consensus layer, network layer, and incentive layer with fewer patterns)?

You can give your opinion on the reasons for this based on your experience, or add what you know about the relevant design patterns. It is recommended to support your opinion with literature.

Examples of aspects that can be considered:

- Do you agree with the algorithms of the consensus layer to be design patterns?
- How do you see a situation where some practical solutions for the network layer and the incentive layer are not identified as design patterns?

Part III. Correctness Evaluation

This part of the question focuses on researchers working on blockchain research within the academic community.

7. What do you think about the relationships between the elements in the decision model? For example, what is reasonable and correct about the relationship?

You can consider the model itself in terms of the correctness of the decision. That is, whether the decision itself is sound and whether the correlations of the elements in the model have been properly analyzed.

Examples of aspects that can be considered:

- Is the mapping relationship between patterns and quality attributes appropriate?
- Do the summarized design patterns effectively address the typical challenges?
- Are each typical challenge and the hierarchy of design patterns reasonable?

Part IV. Usefulness Evaluation

This part of the question focuses on experts with industry experience in blockchain development.

8. For a blockchain application under design and development, can our model help in decision-making? If so, which aspects specifically can it help?

Examples of aspects that can be considered:

- Does the model cover or fit the architectural design needs for the actual development of a blockchain system?
- Does the model help you quickly locate the right design pattern or solution when designing a blockchain system?
- Does the model help architects or developers make good quality decisions and trade-offs before building a blockchain system?

9. For an existing blockchain application, can our model help evaluate the application attributes? If so, what are the specific aspects that can be reflected?

Examples of aspects that can be considered:

- Does the model cover or fit the existing blockchain systems?

- For blockchain systems applying the design model, do the quality attributes analyzed by the model match the attributes of existing systems?
- Does the model effectively help existing systems in quality evaluation?

10. Do you think this decision model is easy to understand and convenient for you to use? And Is it good for a novice to get started?

You can consider the value of this model in terms of its learning cost and user-friendliness for architects or developers with different experiences.

Examples of aspects that can be considered:

- Is the structure diagram of the decision model clear and easy to understand?
- Is the decision model easy to use for developers or architects with different experiences?
- Can the model be quickly learned and used by developers who lack prior knowledge of blockchain or software design patterns?

Part V. Open question

11. Do you have any other suggestions for our work? And are there any other opinions you would like to add?

Questionnaire for the Blockchain-Based Design Pattern Decision Model

Part I. Demographic information

1. What is the scope of your work?

Academic research	☐
Industrial practice	☐
Both of them	☐
Other	Please supplement here

2. How many years have you been working on software architecture?

0-1 year	☐
1-3 years	☐
3-5 years	☐
5-8 years	☐
More than 8 years	☐

3. How many years have you been working on blockchain research?

0-1 year	☐
1-3 years	☐
3-5 years	☐
5-8 years	☐

More than 8 years	☐
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4. What is your current role in the team?

Architect	☐
Development Engineer	☐
Test Engineer	☐
Product Manager	☐
Operation Engineer	☐
Other	Please supplement here

5. To what extent do you know about software engineering, software architecture, and design patterns?

Skilled in theory and related technologies	☐
Knowledge of theory and related technologies in general	☐
Understand basic theoretical concepts and applications	☐
Only heard the name of the technology	☐
Never Heard	☐

6. Are you familiar with the patterns and strategies used in the decision model?

	Familiar with all	Familiar with most of them	Familiar with basics of them	Familiar with few of them	Not familiar at all
Please Choose	☐	☐	☐	☐	☐

7. What benefits do you think the decision model can bring?

Help the system manage quality attributes	☐
Tradeoff quality attributes	☐
Visualize the decisions of different solutions	☐
Decision-making at the time of system architecture can be simulated	☐
Support efficient discovery and evaluation of options and their consequences in a variety of situations	☐
Other	Please supplement here

Part II. Completeness Evaluation

This part of the question focuses on researchers working on blockchain research within the academic community.

8. Is the information in the decision model sufficient to support design pattern decisions? If not sufficient, please indicate.

Please score on a scale of 1-5 and indicate your reasons.

	Completely sufficient5	Mostly sufficient4	Basically sufficient3	Few sufficient2	Not sufficient at all1
Please Choose	☐	☐	☐	☐	☐
Please indicate your reasons					

9. Does the decision model cover important aspects of blockchain system architecture? If any aspect is missing, please indicate

Please score on a scale of 1-5 and indicate your reasons.

	Perfectly complete5	Mostly complete4	Basically complete3	Few complete2	Not complete at all1
Please Choose	☐	☐	☐	☐	☐
Please indicate your reasons					

10. What do you think about the uneven distribution of design patterns across layers (e.g. consensus layer, network layer, and incentive layer with fewer patterns)?

	Reasonable	Debatable	Unreasonable
Please Choose	☐	☐	☐

11. Based on what you know, what do you think are the reasons for the uneven distribution of the layers in the decision model (e.g., less consensus layer, less network layer, less incentive layer)?

The granularity of the selected design patterns varies	☐
The consensus algorithm of the consensus layer is not considered as a design pattern	☐
The design pattern of the consensus layer needs to be further explored and sorted out by the academic community	☐
The network layer does have fewer design patterns at the moment	☐
The experience of incentive mechanism reuse is relatively small, so there are few design patterns in the incentive layer	☐
Other	Please supplement here

Part III. Correctness Evaluation

This part of the question focuses on researchers working on blockchain research within the academic community.

12. Do you think the mapping between patterns and quality attributes in the decision model is correct? If not correct enough, please indicate.

Please score on a scale of 1-5 and indicate your reasons.

	Completely correct5	Mostly correct4	Basically correct3	Few correct2	Not correct at all1
Please Choose	☐	☐	☐	☐	☐
Please indicate your reasons					

13. Do you think the decisions summarized by our decision model can correctly solve the typical problems? If not correct enough, please indicate.

Please score on a scale of 1-5 and indicate your reasons.

	Completely correct5	Mostly correct4	Basically correct3	Few correct2	Not correct at all1
Please Choose	☐	☐	☐	☐	☐
Please indicate your reasons					

Part IV. Usefulness Evaluation

This part of the question focuses on experts with industry experience in blockchain development.

14. When designing the system, do you think our decision model can help you make decisions and tradeoffs? If not support, please indicate.

Please score on a scale of 1-5 and indicate your reasons.

	Completely supported5	Mostly supported4	Basically supported3	Few supported2	Not supported at all1
Please Choose	☐	☐	☐	☐	☐
Please indicate your reasons					

15. For existing systems, do you think our decision model can effectively help you to evaluate the quality? If not support, please indicate.

Please score on a scale of 1-5 and indicate your reasons.

	Completely supported5	Mostly supported4	Basically supported3	Few supported2	Not supported at all1
Please Choose	☐	☐	☐	☐	☐
Please indicate your reasons					

16. Is the decision model easy to understand and use? If not easy to understand and use, please indicate.

Please score on a scale of 1-5 and indicate your reasons.

	Very easy5	Easy4	Neutral3	Difficult2	Very difficult1

Please Choose	☐	☐	☐	☐	☐
Please indicate your reasons					

17. Do you think our decision model is friendly for novices?

	Very friendly5	Mostly friendly4	Basic friendly3	Not very friendly2	Unfriendly1
Please Choose	☐	☐	☐	☐	☐

18. What do you think might be the reasons for not being friendly?

Sufficient prior knowledge and skills are required	☐
No further information on design patterns is provided	☐
Other	Please supplement here

Part V. Open Question

19. Do you have any other suggestions for our work? And are there any other opinions you would like to add?

Please supplement here
